Project Code | *Kod Projek*	**BITU 3973**

UNIVERSITI TEKNIKAL MALAYSIA MELAKA

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

PROJEK SARJANA MUDA 1: PROPOSAL FORM

A TITLE OF PROPOSED PROJECT | *TAJUK PROJEK YANG DICADANGKAN*

AI-ASSISTED CAR RENTAL MANAGEMENT SYSTEM WITH CHATBOT SUPPORT

B DETAILS OF STUDENT | *BUTIRAN PELAJAR*

Name	MUHAMMAD DANIAL HAKIM BIN MOHD SAOFI			**Program:**	BITC [ ] BITD [ ] BITI [ ]
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Semester	2	**Session**	2025/2026	**Email Address**	hdanial211@gmail.com

C PROJECT INFORMATION | *MAKLUMAT PROJEK*

(i) Executive Summary of Project Proposal [Maximum 300 words]

The **AI-Assisted Car Rental Management System with Chatbot Support** aims to modernize and optimize the car rental process using artificial intelligence (AI) and chatbot integration. The system provides customers with a seamless experience for searching, booking, and paying for rental cars, while also offering administrative tools for car fleet management.

This project addresses key challenges faced by car rental businesses, such as fragmented systems, poor customer service, and inefficient vehicle utilization. The AI-based system will provide customers with personalized recommendations and automate booking procedures. Additionally, an integrated chatbot will assist customers with inquiries, booking confirmations, and payment reminders.

The expected outcome is a fully functional, AI-powered car rental system that improves operational efficiency, enhances the customer experience, and simplifies fleet management for businesses.

(ii)

Detailed Proposal of the Project

(a) Introduction *(Project Background and Problem Statements)*

Car rental companies typically face challenges in managing fleets, processing bookings, and handling payments using outdated or disjointed systems. These systems require users to interact with multiple platforms, which leads to inefficiencies and a poor user experience. Moreover, car rental businesses often struggle with optimizing vehicle utilization, which leads to lost revenue.

This project proposes an **AI-Assisted Car Rental Management System** that consolidates all car rental processes into one streamlined platform. It integrates AI and chatbot technology to improve the booking experience, assist customers, and optimize vehicle utilization, ultimately benefiting both businesses and customers.

(b) Objectives of the Project

This project aims to achieve the following objectives:

1. ***To Analyze** the current challenges in car rental management systems and identify areas for improvement.*
2. ***To Develop** an AI-powered system with integrated chatbot support to handle customer inquiries and bookings.*
3. ***To Evaluate** the system's effectiveness, usability, and AI features through comprehensive testing.*

(c) Scope of the Project

The scope of this project includes:

1. **Customer Booking Portal**: A user-friendly interface for customers to search for available cars, make reservations, and complete payments securely.
2. **AI Chatbot**: A chatbot integrated with the system to assist customers with booking inquiries, payment reminders, and troubleshooting.
3. **Admin Panel**: A backend interface for administrators to add cars, manage bookings, track payments, and generate business reports.
4. **Database Management**: Organizing and storing car inventory, customer profiles, booking details, and payment records.
5. **Mobile Responsiveness**: Ensuring that the system is accessible on mobile devices for convenient on-the-go booking.

(d) Expected Outcome/ Proposed Solution

The expected outcome of this project is the creation of a fully functional **Car Rental Management System** that offers:

- A **customer portal** with features like booking, car search, and payment integration.
- An **AI-powered chatbot** to improve customer interaction and automate booking processes.
- A **user-friendly admin panel** to simplify the management of car fleets and bookings.
- **Real-time updates** on car availability, booking statuses, and payment confirmations.
- That **can be expanded** with features such as customer reviews, and advanced AI capabilities.

D REFERENCES | *RUJUKAN*

State your references (Minimum 10 references)

1	***Adamopoulou, E., & Moussiades, L. (2020).*** *An overview of chatbot technology. IFIP Advances in Information and Communication Technology, 584, 373–383. https://doi.org/10.1007/978-3-030-49186-4_31*
2	***Gupta, R. (2024).*** *AI-driven fleet analytics: Revolutionizing modern fleet management. ResearchGate, 1-15. https://www.researchgate.net/publication/390194424_AI-DRIVEN_FLEET_ANALYTICS_REVOLUTIONIZING_MODERN_FLEET_MANAGEMENT*
3	***Duong, T., Pham, Q., Oh, J., & Do, A. (2025).*** *Can AI chatbot adoption bridge the gap between intention and e-booking behavior? Sustainability, https://www.mdpi.com/2071-1050/17/17/7673*
4	***Nguyen, Q. H., & Do, T. K. (2026).*** *The impact of AI chatbot on customer willingness to pay: An empirical investigation. Journal of Open Innovation: Technology, Market, and Complexity? https://www.mdpi.com/2673-5768/7/3/68*
5	***Zhang, L., & Wang, H. (2026).*** *Mapping the research landscape of AI chatbot adoption in tourism and hospitality. Journal of Hospitality and Tourism Management, https://www.sciencedirect.com/org/science/article/pii/S1947820826000048#:~:text=AI%20chatbots%20are%20widely%20used,chatbots%20still%20encounter%20notable%20challenges.*
6	***Kumar, S., & Singh, P. (2025).*** *Advanced car rental system: Integrating blockchain, AI and real-time inventory management for enhanced user experience. ResearchGate, 1-12. https://www.researchgate.net/publication/394119741_Advanced_Car_Rental_System_Integrating_Blockchain_AI_and_Real-Time_Inventory_Management_for_Enhanced_User_Experience*
7	***Ali, M., & Rahman, S. (2025).*** *Driving consumer engagement through AI chatbot experience. IEEE Xplore, https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11088190*
8	***Rybo AI. (2024).*** *Transforming car rentals with AI chatbots: Benefits and use cases. Retrieved April 23, 2026, from https://www.rybo.ai/car-rental-chatbot/*
9	***Agentive AI. (2024).*** *7 best smart AI agent systems for car rental management in 2024. Retrieved April 23, 2026, from https://agentiveaiq.com/listicles/7-best-smart-ai-agent-systems-for-car-rental*
10	***AirentoSoft. (2024).*** *All-in-one car rental software for fleet management and online booking. Retrieved April 23, 2026, from https://airentosoft.com/car-rental-software*

E DECLARATION BY STUDENT | *AKUAN PELAJAR*

(i) **Date:** 9 April 2026 **Student's Signature:**

E RECOMMENDED BY SUPERVISOR
PERAKUAN OLEH PENYELIA

Recommended [/]

Not Recommended [ ]

Comments:

For the next revision, please update the project title and scope to focus on the core car rental management system first. The AI chatbot component should be planned for future add on. The proposal is ready for approval

Supervisor's Name:

MUHAMMAD FAIZ BIN SUPIAN

Signature & Stamp:

MUHAMMAD FAIZ BIN SUPIAN
Pensyarah
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Date: 22/5/2026